

Delphes vs. CMS for $HH \rightarrow b\bar{b}\tau\tau$: where we stand, and one decision to take

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Abstract

The $\tau_h\tau_h$ yield deficit that motivated this work (a factor ~ 5) is closed: the cutflow agrees with CMS at every stage (1.01) and eight of ten NSBI features agree. The two that do not, $m_{\tau\tau}$ and $\Delta R_{\tau\tau}$, are both on the τ side. The $m_{\tau\tau}$ offset is now traced: it is in the τ four-vectors, fakes are excluded, and the Delphes τ energy response carries a one-sided tail CMS does not have — at gen $p_T = 20\text{--}30$ GeV, one Delphes τ in ten has more than twice the median response, against 1.08 on CMS. A multiplicative median map cannot reshape a distribution, so the current correction cannot close this. That leaves three options, priced in Section 2; the measurements are in Section 3.

1 Status

Stock CMS card plus the patches in `docs/card_patch_validation.md`. All tuning is *downstream*, so samples are never re-made; every map is measured on a private 2024 NanoAODv15 anchor, so “tuned” means “measured on CMS and transferred”.

quantity	status	how
b-tag efficiency (b , c , light)	✓ tuned	stochastic re-tag from <code>Jet.Flavor</code> at anchor UParT-AK4 Medium
b-jet energy scale	✓ tuned	inverted to unity against <code>GenJet</code> (self-anchored)
m_{bb} peak and width	✓ matched	follows
τ_h ID efficiency	✓ tuned	anchor <code>GenVisTau</code> → <code>DeepTau</code> v2.5 Medium; includes HPS reco inefficiency
τ_h visible mass	✓ tuned	sampled per jet from anchor per- p_T quantiles
τ_h acceptance, jet- τ overlap	✓ identical	$p_T > 20$, $ \eta < 2.3$; pair first, then $\Delta R > 0.4$
p_T^{miss} resolution	✓ tuned	Gaussian smearing, 16.5 → 32.6 GeV/component
lepton reco efficiency	✓ tuned	per- p_T SFs ($1.57 \rightarrow 1.22$ e ; $1.30 \rightarrow 1.13$ μ)
di- τ mass estimator	✓ identical	same covariance-free FastMTT both sides
τ_h energy scale	✗ tuned	CMS/Delphes response ratio — under-corrects, Section 3.1

Table 1: Object status. Every map is derived on the anchor and applied downstream; none needs a card change.

✓ agree	m_{HH} (the κ_λ observable, across $\kappa_\lambda = 0, 1, 5$); m_{bb} (both peak ~ 112 GeV); p_T^{HH} ; $\Delta\phi_{HH}$; p_T^{H1} , p_T^{H2} ; ΔR_{bb} (a
✗ disagree	$m_{\tau\tau}$ (~ 7 GeV high, Delphes broader); $\Delta R_{\tau\tau}$ (excess at 2.2–3.2)

Table 2: NSBI inputs at $\kappa_\lambda = 0$; other κ_λ points behave the same. Yields: gen-matched cutflow 1.01, analysis-level $1.18\times$.

Known limitations, accepted. None is implicated in the two disagreements, but they bound what the samples support. **(i)** `tau_mistag` is measured on the b -rich signal anchor; fake rates are strongly flavour-dependent, so it is *not* valid for the fake-dominated backgrounds (QCD, W+jets), which need their own measurement. **(ii)** p_T^{miss} smearing is isotropic and flat where the real resolution is anisotropic and grows with ΣE_T . **(iii)** No pileup: jet energy *resolution* is untuned and pileup jets cannot exist; both act on the $b\bar{b}$ side, which agrees. **(iv)** The sampled τ_h mass is drawn independently of the τ energy and the partner leg (unmeasured). **(v)** FastMTT is covariance-free — run identically both sides, so it limits absolute $m_{\tau\tau}$ resolution, not agreement. **(vi)** Two defects in the energy-scale *application*: fake τ_h match neither branch of `escale_factor` and stay uncorrected, and the τ branch matches all gen τ (`|pid|==15`) including leptonic. **(vii)** Every map is flat-extrapolated above 300 GeV — untested, and κ_λ discrimination lives at high m_{HH} , hence high jet p_T .

2 The decision

A CMS τ_h is an HPS object built from a narrow set of decay products; a Delphes τ_h is an $R = 0.4$ jet carrying `TauTag`. Mass, energy and direction are therefore wrong at source and we correct the first two rather than fix the representation. Section 3.1 shows the correction cannot succeed as built: a multiplicative per- p_T median cannot depopulate a kinematically forbidden region, and the Delphes response is one-sidedly broader at matched median by $3\times$ at low p_T . That is narrower than “only rebuilding works” — hence three options, not two.

(A) Rebuild the τ_h from particle flow. The card already writes the three `EFlow` collections, per its header, precisely for “ τ redefinition”. HPS-like: seed from a jet, collect charged hadrons and photons in a narrow signal cone (fixed $\Delta R \simeq 0.1$ or p_T -dependent $3.0/p_T$ bounded at 0.05–0.1), form the standard decay-mode hypotheses, sum signal-cone constituents, isolation annulus as a fake handle. Mass, energy and direction become correct *by construction*; for a pheno paper this removes a fast-simulation limitation instead of documenting one. Weeks, but *not* on the critical path — the multi-day item is running the full 5×10^5 -event samples and training the NSBI.

Against: it cannot reproduce DeepTau, so the ID and fake-rate maps remain (they work). It is not obviously the cure for $\Delta R_{\tau\tau}$ — CMS HPS also seeds from AK4 jets, so merging scales are similar. *The caveat we most want checked:* Delphes neutral PF candidates are *calorimeter towers* ($\sim 0.02 \times 0.02$ barrel), not clustered PF objects. Since π^0 content sets both the visible mass and the decay mode, an HPS-like reconstruction here would be approximate in exactly the place that matters — possibly no better than the anchor-sampled mass already in use.

(B) Correct the distribution, not the scale. What is forbidden is a multiplicative *median* map, not every downstream correction. For each gen-matched τ -jet set $p_T^{\text{reco}} = p_T^{\text{gen}} \times r$ with r drawn from the anchor’s response quantiles — the machinery already used for the visible mass, applied to the energy. Reproduces the CMS response distribution including its tail. About a day. *Against:* only covers the $\sim 96\%$ gen-matched (fakes have no gen τ); p_T^{miss} is built from reco objects, so resampling afterwards breaks the balance (we already smear p_T^{miss} independently, so a difference of degree, but it compounds); the τ energy becomes partly synthetic, which for an unbinned NSBI result resting on realistic simulation is defensible but weaker than a rebuild; and it does nothing for $\Delta R_{\tau\tau}$, decay mode or charge.

(C) Accept it and document. Eight of ten features and the yield already agree. The $m_{\tau\tau}$ residual is ~ 7 GeV on ~ 100 , and NSBI trains on shapes with a fitted normalisation.

What does *not* decide it. Rebuilding is sometimes said to retire three corrections — visible mass, energy scale, visible- τ reference. Badly weighted: the mass map’s residual imperfection

is worth 0.11% on $\langle m_{\tau\tau} \rangle$, so retiring it buys almost nothing, and the visible- τ reference is a derivation-side simplification. One of the three matters, and it is the energy scale.

Questions for τ expertise.

1. Is tower-granular neutral reconstruction adequate for a useful decay-mode assignment, or does it degrade π^0 content enough that the rebuilt mass is no better than the anchor-sampled one?
2. Signal cone: fixed 0.1 or p_T -dependent, over the 20–60 GeV range that dominates here?
3. Seeding: jets (as HPS) or direct PF clustering? Would either change the efficiency for collimated pairs, $\Delta R_{\tau\tau} \lesssim 1$?
4. Is the $\Delta R_{\tau\tau}$ pattern — over-efficient above 2.5, possibly under-efficient below 1 — recognisable as a jet-versus-HPS effect?
5. Is option (B) acceptable in a pheno paper, or does resampling a reconstructed quantity undercut the claim?
6. Is a 1.18 $\times$ analysis-level yield excess acceptable for unbinned NSBI, where training uses shapes and normalisation is fitted?

3 The two open questions

3.1 $m_{\tau\tau}$

$m_{\tau\tau} = m_{\text{vis}}/\sqrt{x_1 x_2}$ factorises, so the *visible* di- τ mass m_{vis} — the two τ four-vectors, no p_T^{miss} , no FastMTT — separates a τ scale error from an x -fit error. The two have disjoint fixes.

It is the four-vectors. m_{vis} is offset by *at least* as much as $m_{\tau\tau}$ (peak ratios ≈ 1.24 vs ≈ 1.17), so FastMTT is not the source; if anything it compresses. Both legs are individually harder. With an applied scale of $\simeq 0.85$ and a residual $\simeq 1.24$, the raw Delphes τ are $\sim 46\%$ harder than CMS while the map corrects 15%.

Not fakes. Splitting gen-matched from fake gives fake fractions 4.1% Delphes vs 4.3% CMS with indistinguishable shapes; the gen-matched component carries everything — the peak displacement and an unphysical tail at $m_{\text{vis}} > 125$ GeV, kinematically impossible for $H \rightarrow \tau\tau$ and absent on CMS. Same pattern as $\Delta R_{\tau\tau}$.

The response distribution. Plotting the response (reco τ p_T / gen-visible- τ p_T) as a *distribution*, each side divided by its own median so the scale is removed by construction, leaves only what the map cannot change. One control first: without a τ ID, Delphes matches a gen τ to the nearest jet of *any* kind (so a lost τ -jet can match a b -jet), while CMS matches within the Tau collection — and mis-matching mimics the same p_T dependence. With the ID required on both sides the tail survives (Table 3, Figure ??):

gen p_T [GeV]	median		q_{95}/median		ratio
	Delphes	CMS	Delphes	CMS	
20–30	1.184	0.989	3.378	1.135	3.0 $\times$
30–50	1.114	0.987	1.873	1.105	1.7 $\times$
50–100	1.066	0.978	1.325	1.102	1.2 $\times$
100–300	1.035	0.970	1.195	1.115	1.07 $\times$

Table 3: τ_h energy response, τ ID required both sides. q_{95}/median is pure *shape*: the scale is divided out.

[Figure omitted: `figures/tau_response_id.pdf` not present. Run `tau_response.py` with `--require-id` and copy its output there.]

CMS is flat in p_T (1.10–1.14), as a calibrated object with constant fractional resolution should be ($q_{90}/\text{median} = 1.076 \Rightarrow \sigma \simeq 6\%$, right for HPS), so the anchor is sound. Delphes is not flat and converges on CMS as p_T rises — what a constant *absolute* energy excess gives, large as a fraction at 25 GeV and small at 175. Medians converge alongside, as one cause predicts. Sharpest number: at 20–30 GeV the Delphes q_{90}/median is 2.156 against 1.076 on CMS. That is not a resolution. One caveat survives: **TauTag** is a stochastic draw uncorrelated with contamination while DeepTau actively rejects it, so part of the gap is a *selection* correlation — equally unreachable by a scale map, so it does not change Section 2, but the tail is not purely about cone size.

Excluded. (i) *The sampled mass.* The FastMTT floor $x_{\min} = (m_{\text{vis}}^{\text{leg}}/m_{\tau})^2$ exceeds a typical fitted $x \simeq 0.5\text{--}0.9$ only above ~ 1.26 GeV, so below that the mass is inert: displacing the whole low-mass region by 200 MeV moves $\langle m_{\tau\tau} \rangle$ by 0.00%, against -2.4% in the a_1 tail. Replacing the 21-quantile grid with an atom at $m_{\pi\pm}$ plus a histogram CDF changes it by 0.11%. (ii) *A symmetric resolution mismatch.* $7\% \rightarrow 25\%$ at fixed median moves the mean 0.05% (RMS $+2\%$) — but that test used a symmetric log-normal, so it does *not* cover the one-sided tail above.

3.2 $\Delta R_{\tau\tau}$: a selection effect, cause unknown

Gen-level $\Delta R_{\tau\tau}$ spectra agree, so the samples are the same physics; the efficiencies differ — Delphes selects 0.22–0.23 at gen $\Delta R > 2.5$ where CMS selects 0.15–0.17 ($\sim 3\sigma$), with a weaker ($\sim 2\sigma$) hint of loss below $\Delta R < 1$. Both push the distribution rightward. Excluded by measurement: fakes (the discrepancy is entirely gen-matched); τ -pair selection (symmetrising moved $\Delta R_{\tau\tau}$, not $m_{\tau\tau}$); CMS τ miscalibration (anchor response 0.97–0.99). This matters more than its size: it is a *selection* effect, so no downstream correction reaches it, and unlike $m_{\tau\tau}$ we have no candidate mechanism.

A Reference

All reproducible from `delphes-pipeline`.

1. **Cutflow** (`tauh_cutflow.py`). Five stages — gen visible τ_h , reco, ID, cleaned jets, FastMTT — walked identically both sides on *conditional* efficiencies. FastMTT once carried the whole deficit (12% vs 99%); now 1.01 overall. Tests inject a loss at one stage and assert it is attributed there only.
2. **FastMTT, and why the mass is sampled.** The hadronic prior vanishes unless $x_{\min} \leq 1$; an AK4 jet mass (5–20 GeV vs $m_{\tau} = 1.777$) admits no solution, returning NaN and silently removing $\sim 89\%$ of pairs. The anchor mass fixes it, but must be *sampled*: it is a one-sided floor, so a single median relaxes a real constraint for every leg above it, inflating the width $\sim 60\%$ and mean $\sim 9\%$ while leaving the *mode* unchanged — a mode-gated $Z \rightarrow \tau\tau$ candle would miss it.
3. **The reference error, twice.** A GenJet is also a jet and also contains UE, so it is not the analogue of GenVisTau. Against a true visible τ ($p_{\tau} - p_{\nu_{\tau}}$, exact for one neutrino) the Delphes τ -jet is 17% harder at 25 GeV, $\sim 2\%$ at 250. The same error appeared in the escale map and in the cutflow’s stage 1.
4. **Order of application.** Maps measured at CMS scale must be applied at CMS scale; tagging raw jets evaluated a steep efficiency $\sim 17\%$ too far right. Applying the scale first moved the excess $1.25 \rightarrow 1.18$.
5. **Gen- τ check** (`gen_tau_check.py`). Our visible- τ construction vs CMS GenVisTau on the same events: efficiency 0.991, purity 0.998, stage-1 rate to $\sim 1.6\%$.
6. **m_{vis} and the split** (`nsbi_overlay.py --diagnostics --split-gen-matched`). m_{vis} uses neither p_T^{miss} nor the fit, so it attributes an offset to one of two disjoint causes; tests pin $m_{\text{vis}} \leq m_{\tau\tau}$, invariance under $p_T^{\text{miss}} 40 \rightarrow 200$ GeV, and exact linearity in the τ scale. The split normalises to each side’s total, so the fake curve’s area is the fake fraction.

7. **Response shape** (`tau_response.py`). Distribution per $\text{gen-}p_T$ bin, lower row divided by each side's own median. Tests assert the separation: a pure scale difference vanishes under normalisation, an injected one-sided tail survives. `--gate` and `--require-id` expose two asymmetries in the shipped derivation (anchor gates on gen and conditions on DeepTau; Delphes gates on reco and conditions on nothing).
8. **Map accuracy**. No map is statistics-limited at 2×10^5 anchor events (b -tag 0.25–0.42%/bin, τ_h eff 0.43–1.26%, escale 0.2–0.3%, p_T^{miss} 0.16–0.21%); what limits a map is np , not n . Two defects do not improve with statistics: the mass quantile grid interpolates across the empty region between the $\pi^\pm$ atom and the ρ shoulder ($\sim 6\times$ over-populated, measured inert), and the 300 GeV flat extrapolation (untested).
9. **Generator copies**. Delphes writes the full Pythia history, CMS **GenPart** is pruned. Rule: anything that **counts** generator objects must collapse copies first; anything that **matches** geometrically need not.

Reproducing.

```
# --max-events is REQUIRED: derive_maps caps the anchor from the config but reads
# the Delphes side uncapped.
python -m delphes_pipeline.tuning.derive_maps --config config.v1.yml \
    --output cards/tuning/maps_v1.json --max-events 200000
python -m delphes_pipeline.validation.run_validation --config config.v1.yml
python scripts/tauh_cutflow.py --config config.v1.yml [...] --gen-dr plots/gen_dr
python scripts/nsbi_overlay.py --config config.v1.yml [...] --tautau-only \
    --diagnostics --split-gen-matched
python scripts/tau_response.py --config config.v1.yml [...] --require-id
```